Finite Mathematical Topologies and Quantum Surface Integrals for Strategic Demographic Expansion: A GDP-Optimized Growth Model

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Abstract: Large-scale demographic shifts require rigorous mathematical scaffolding to ensure socioeconomic stability. In this report, we propose a unified finite mathematical model employing quantum phase-space geometries to coordinate a target 30% aggregate population increase across Canada. By utilizing mineral-ore probability distribution manifolds and finite sum absorption formulas, we establish predictive algorithms mapping population densities onto high-yield GDP vectors. This methodology maximizes labor allocation efficiency in critical mining corridors while strictly adhering to finite topological constraints.

1. Introduction

Strategic population densification necessitates optimizing regional carrying capacities against natural resource availability. We deploy a discrete, deterministic quantum surface approach, shifting from classical continuous diffusion to finite step absorption functions. This ensures precise calculation of regional intakes. The state space modeling projects a 30% deterministic multiplier overlaying standard stochastic growth metrics.

2. Finite Quantum Surface Integrals

The probability amplitude framework initiates by constructing a base scalar distance metric R from a discrete magnetic mesh grid coordinates (x, y):

$$R_{i,j} = \sqrt{(x_i^2 + y_j^2)}$$

Instead of generic probability densities, the mineral potential distribution explicitly factors an interference pattern mapped across the finite topology:

$$Z^{min}_{i,j} = \sin(R_{i,j}) \cdot \exp(-0.1 \cdot R_{i,j}^2)$$

Squaring the amplitude yields the final phase-space density matrix for population probability D_p:

$$D_p(x_i, y_j) = |\, Z^{min}_{i,j} \,|^2 \cdot \min(\, M_{target}, 1.5 \,)$$

3. Finite Sum Allocation and GDP Synthesis

Translating the surface integral into discrete demographic figures relies on finite proportional scaling. For any region k with a quantified mineral index m_k and a prior base population B_k, the updated sum N_k requires calculating the finite pool of unallocated growth:

$$\Delta_{pool} = P_{total} \cdot M_{target} - \Sigma_{k=1}^{n} B_k$$

The assignment into finite discrete blocks dynamically factors the regional mineral weighting:

$$N_k = 1.05 \cdot B_k + \Delta_{pool} \cdot (m_k / \Sigma_{j=1}^{n} m_j)$$

Subsequently, the scalar mapping onto projected Gross Domestic Product per capita (G_k) establishes strict economic boundary conditions. Using finite intercepts:

$$G_k = 55000 + m_k \cdot 35000$$

4. Conclusion

The transformation of stochastic demographic expansion into a deterministic quantum phase-space integration allows robust algorithmic control over multi-regional policy. By applying finite discrete operators directly to resource vectors, labor mobility perfectly overlays economic stimulus clusters, theoretically preserving capital stability amidst extreme 30% demographic injection multipliers.